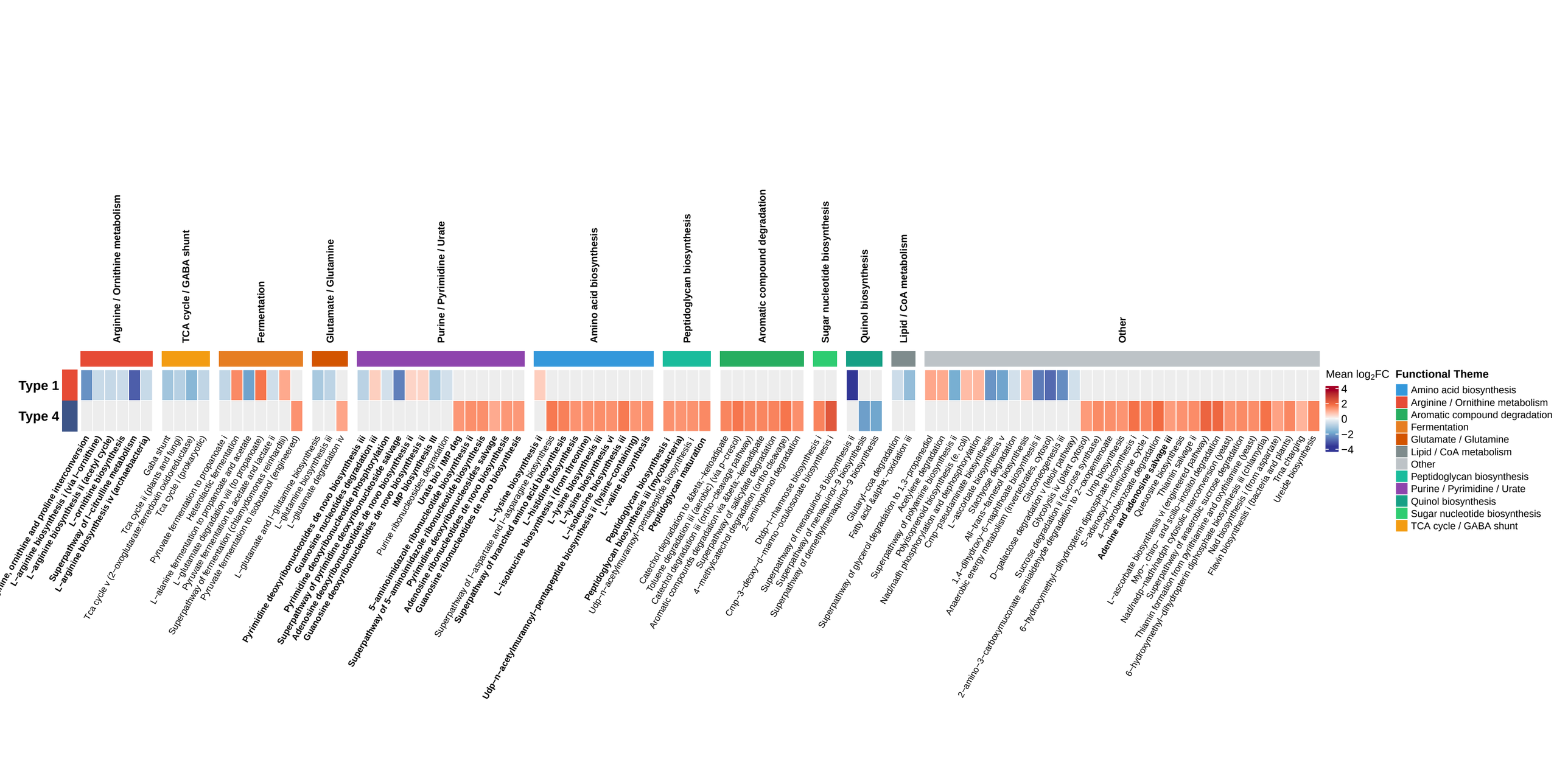
Arginine / Ornithine metabolism
TCA cycle / GABA shunt
Fermentation
Glutamate / Glutamine
Purine / Pyrimidine / Urate
Amino acid biosynthesis
Peptidoglycan biosynthesis
Aromatic compound degradation
Sugar nucleotide biosynthesis
Quinol biosynthesis
Lipid / CoA metabolism
Other
Type 1
Type 4
Mean log2FC
4
2
0
−2
−4
Functional Theme
Amino acid biosynthesis
Arginine / Ornithine metabolism
Aromatic compound degradation
Fermentation
Glutamate / Glutamine
Lipid / CoA metabolism
Other
Peptidoglycan biosynthesis
Purine / Pyrimidine / Urate
Quinol biosynthesis
Sugar nucleotide biosynthesis
TCA cycle / GABA shunt
L−arginine, ornithine and proline interconversion
L−arginine biosynthesis i (via L−ornithine)
L−arginine biosynthesis ii (acetyl cycle)
Superpathway of L−ornithine biosynthesis
L−arginine biosynthesis iv (archaebacteria)
Tca cycle v (2−oxoglutarate:ferredoxin oxidoreductase)
Gaba shunt
Tca cycle ii (plants and fungi)
Tca cycle i (prokaryotic)
Pyruvate fermentation to propanoate i
Heterolactic fermentation
L−alanine fermentation to propanoate and acetate
L−glutamate degradation viii (to propanoate)
Superpathway of fermentation to acetate and lactate ii
Pyruvate fermentation (chlamydomonas reinhardtii)
Pyruvate fermentation to isobutanol (engineered)
L−glutamate and L−glutamine biosynthesis
L−glutamine biosynthesis iii
L−glutamate degradation iv
Pyrimidine deoxyribonucleotides de novo biosynthesis iii
Pyrimidine deoxyribonucleotides degradation iii
Guanosine nucleotides degradation iii
Superpathway of pyrimidine deoxyribonucleotide phosphorylation
Pyrimidine deoxyribonucleotides de novo biosynthesis ii
Guanosine nucleotides de novo biosynthesis ii
Adenosine deoxyribonucleotides de novo biosynthesis ii
Guanosine deoxyribonucleotides de novo biosynthesis ii
Adenosine nucleotides degradation iii
Superpathway of 5−aminoimidazole ribonucleotide biosynthesis
IMP biosynthesis ii
5−aminoimidazole ribonucleotide biosynthesis iii
Purine ribonucleosides degradation
Urate bio / IMP deg
5−aminoimidazole ribonucleotide biosynthesis ii
Pyrimidine deoxyribonucleotide biosynthesis
Adenosine ribonucleotide deoxyribonucleotides salvage
Guanosine ribonucleotides de novo biosynthesis
Superpathway of ribonucleotides de novo biosynthesis
Superpathway of l−aspartate and l−asparagine biosynthesis
Superpathway of branched amino acid biosynthesis
L−isoleucine biosynthesis ii
Udp−n−acetylmuramoyl−pentapeptide biosynthesis ii (lysine−containing)
L−lysine biosynthesis ii
L−lysine biosynthesis i
L−histidine biosynthesis i (from threonine)
L−lysine biosynthesis iii
L−isoleucine biosynthesis iii
L−lysine biosynthesis vi
L−valine biosynthesis
Peptidoglycan biosynthesis i (mycobacteria)
Peptidoglycan biosynthesis iii (mycobacteria)
Udp−n−acetylmuramoyl−pentapeptide biosynthesis i
Peptidoglycan maturation
Catechol degradation to β−ketoadipate
Toluene degradation iii (aerobic) (via p−cresol)
Catechol degradation iii (ortho−cleavage pathway)
Aromatic compounds degradation via β−ketoadipate
Superpathway of salicylate degradation
4−methylcatechol degradation (ortho cleavage)
2−aminophenol degradation
Cmp−3−deoxy−d−manno−octulosonate biosynthesis i
Ddqh−l−rhamnose biosynthesis i
Superpathway of menaquinol−8 biosynthesis ii
Superpathway of menaquinol−9 biosynthesis
Superpathway of demethylmenaquinol−9 biosynthesis
Glutaryl−coa degradation
Fatty acid α−oxidation iii
Superpathway of glycerol degradation to 1,3−propanediol
Acetylene degradation
Superpathway of polyamine biosynthesis ii
Polyisoprenoid biosynthesis (e. coli)
Nadh/nadh phosphorylation and dephosphorylation
Cmp−pseudaminate biosynthesis
L−ascorbate biosynthesis v
Stachyose degradation
Alt−trans−farnesol biosynthesis
1,4−dihydroxy−6−naphthoate biosynthesis ii
Anaerobic energy metabolism (invertebrates, cytosol)
D−galactose degradation v (leloir pathway)
Gluconeogenesis iii
Sucrose degradation ii (sucrose synthase)
Glycolysis iv (plant cytosol)
2−amino−3−carboxymuconate semialdehyde degradation to 2−oxopentenoate
6−hydroxymethyl−dihydropterin diphosphate biosynthesis i
S−adenosyl−l−methionine cycle i
4−chlorobenzoate degradation
Adenine and adenosine salvage iii
Queuosine biosynthesis
Thiamin salvage ii
L−ascorbate biosynthesis vi (engineered pathway)
Myo−, chiro− and scyllo−inositol degradation
Nadh/nadp−nadh/nadph cytosolic interconversion (yeast)
Superpathway of anaerobic sucrose degradation
Thiamin formation from pyrithiamine and oxythiamine (yeast)
6−hydroxymethyl−dihydropterin diphosphate biosynthesis iii (chlamydia)
Nad biosynthesis i (from aspartate)
Flavin biosynthesis i (bacteria and plants)
Trna charging
Ureide biosynthesis